



INNOVA JUNIOR COLLEGE
JC 2 PRELIMINARY EXAMINATION
in preparation for General Certificate of Education Advanced Level
Higher 2

CANDIDATE
NAME

CLASS

INDEX NUMBER

MATHEMATICS

9758/02

Paper 2

11 September 2018

3 hours

Additional Materials:

Answer Paper
Cover Page
List of Formulae (MF 26)

READ THESE INSTRUCTIONS FIRST

Do not open this booklet until you are told to do so.

Write your name, class and index number on the work you hand in.
Write in dark blue or black pen on both sides of the paper.
You may use an HB pencil for any diagrams or graphs.
Do not use staples, paper clips, glue or correction fluid.

Answer **all** the questions.

Give non-exact numerical answers correct to 3 significant figures, or 1 decimal place in the case of angles in degrees, unless a different level of accuracy is specified in the question.

You are expected to use an approved graphing calculator.

Unsupported answers from a graphing calculator are allowed unless a question specifically states otherwise.

Where unsupported answers from a graphing calculator are not allowed in a question, you are required to present the mathematical steps using mathematical notations and not calculator commands.

You are reminded of the need for clear presentation in your answers.

At the end of the examination, fasten all your work securely together.

The number of marks is given in brackets [] at the end of each question or part question.

This document consists of **7** printed pages and **1** blank page.



Section A: Pure Mathematics [40 marks]

1 It is given that

$$f(x) = \begin{cases} 2x-8 & \text{for } 0 \leq x < 2, \\ -x-2 & \text{for } 2 \leq x < 6, \end{cases}$$

and that $f(x) = f(x+6)$ for all real values of x .

- (i) Evaluate $f(-21) + f(49)$. [2]
- (ii) Sketch the graph of $y = f(x)$ for $-6 \leq x \leq 10$. [3]
- (iii) Find $\int_{-2}^7 f(x) \, dx$. [3]

2 The function f is defined by

$$f(x) = \cos x - \sqrt{3} \sin x, \text{ where } -\frac{\pi}{3} \leq x \leq \frac{2\pi}{3}.$$

- (i) Express $f(x)$ in the form $R \cos(x + \alpha)$ where R and α are exact positive constants to be found. State the range of f . [3]
- (ii) Show that $f(x)$ decreases as x increases. [2]
- (iii) Define f^{-1} . [2]
- (iv) The composite function fg is defined by

$$fg(x) = x - 3, \text{ where } 1 \leq x \leq 3.$$

Find $g(x)$. [2]

- 3** Liquid is being poured into a cylindrical tank at a constant rate of $1200 \text{ cm}^3 \text{ s}^{-1}$ and is leaking out of a hole in the base at a rate proportional to the square root of the height of liquid in the tank. The tank is initially empty, and at time t seconds after pouring, the liquid in the tank has volume $V \text{ cm}^3$ and height $h \text{ cm}$. The circular surface area of the liquid in the tank is 3000 cm^2 .

- (i) Write down a differential equation expressing $\frac{dV}{dt}$ in terms of h . Hence show that $\frac{dh}{dt} = 0.4 - k\sqrt{h}$, where k is a positive constant. [3]

When $h = 36$, liquid is leaking out of the hole at $360 \text{ cm}^3 \text{ s}^{-1}$.

- (ii) Show that $k = 0.02$. [1]
- (iii) By using the substitution $\sqrt{h} = 20 - x$, find the particular solution of t in terms of h . [5]
- (iv) Hence find the time taken for the liquid to reach a height of 100 cm , giving your answer in minutes and seconds, correct to the nearest second. [2]

- 4** Given that $f(x) = \frac{1}{\sqrt{1+x^2}}$ and $g(x) = \frac{a}{\sqrt{1+(0.5x-1)^2}}$ where a is a constant greater than 2, describe fully a sequence of transformations which would transform the graph of $y = f(x)$ onto the graph of $y = g(x)$. [3]

The region R is bounded by the curves $y = f(x)$, $y = g(x)$, the line $x = 2$ and the y -axis. A sculpture is made in the shape of the solid of revolution formed by rotating R through 2π radians about the x -axis.

- (i) Find the exact volume of the sculpture, giving your answer in terms of a and π . [4]
- (ii) Another region S is bounded by the curve $y = f(x)$, the line $x = 2$ and the x - and y - axes. A second sculpture takes the shape of the solid of revolution formed by rotating S through 2π radians about the y -axis. Find the exact volume of the second sculpture. [3]
- (iii) Given that the volume of the first sculpture found in part (i) is at least 50 times the volume of the second sculpture found in part (ii), find the smallest integer value of a . [2]

Section B: Probability and Statistics [60 marks]

- 5 The random variable X has the distribution $B(25, p)$, where $0 < p < 1$. Given that $P(X \leq 1) = 0.15$, write down an equation for the value of p and find this value numerically. Hence find $\text{Var}(X)$. [4]

- 6 Find the number of ways in which the letters of the word APPRECIATE can be arranged if

- (i) vowels (A, E, I) and consonants (P, R, C, T) must alternate, [2]
- (ii) between the two Es, there are exactly two other letters and at least one of which must be an A. [3]

- 7 The discrete random variable X takes values 0, 1, 2 and 3 only. The probability distribution of X is shown in the table, where p is a constant and $0 < p < \frac{1}{10}$.

x	0	1	2	3
$P(X = x)$	$1 - 6p$	$3p$	$2p$	p

- (i) Given that $\text{Var}(X) = 0.75$, find the value of $E(X)$. [3]
 - (ii) The random variable S is the sum of n independent observations of X , where n is large. Given that the probability that S exceeds 150 is at least 0.75, find the set of possible values of n . [3]
- 8 For events A , B and C , it is given that $P(A) = 0.5$, $P(B) = 0.45$ and $P(C) = 0.35$. It is further given that $P(B|C) = 0.5$, $P(A \cap C) = 0.15$ and $P(A \cap B \cap C) = 0.1$.
- (i) Find $P(A' \cap B \cap C)$. [2]
 - (ii) Given also that events A and B are independent, find $P(A \cup B)$. [2]
 - (iii) Given instead that events A and B are **not** independent, find the greatest and least possible values of $P(A' \cap B' \cap C')$. [4]

- 9** Bottles of tomato juice produced by a company are said to contain 250 ml, with a standard deviation of 10 ml. After receiving feedback from some consumers regarding the volume of tomato juice per bottle, the manager takes a random sample of 50 bottles to test whether the mean volume has been overstated. He measures the volume, x ml of tomato juice in each bottle and the sample mean volume is found to be 247.5 ml.

- (i) State appropriate hypotheses for the test, defining any symbols you use. [2]
- (ii) Find the p -value of the test and state the meaning of this p -value in context. [2]
- (iii) State, giving a reason, whether it is necessary to assume a normal distribution for this test to be valid. [1]

The company installs a new machine to produce smaller bottles of tomato juice with mean volume μ_0 ml. A random sample of these smaller bottles of tomato juice is taken. The sample size is 60 and the volumes, y ml, are summarised as follows.

$$\sum y = 10\,757 \qquad \sum y^2 = 1\,931\,597$$

- (iv) Calculate unbiased estimates of the population mean and variance of the volume of smaller bottles of tomato juice. [2]
- (v) A two-tail test is to be carried out at the 5% significance level by the manager. Find the range of values of μ_0 , correct to 1 decimal place, such that the null hypothesis will not be rejected. [3]

- 10** A company wants to investigate the effect of using strong acid solution in reducing the weight of metal plates. Eight metal plates are randomly selected and immersed in a strong acid solution for different lengths of time, t hours. The percentages of weight loss, w %, are calculated and the results are shown in the table below.

t	100	150	200	250	300	350	400	450
w	0.80	1.40	2.00	2.31	2.53	2.65	2.71	2.77

- (i) Calculate the product moment correlation coefficient between t and w , and explain whether your answer suggests that a linear model is appropriate. [3]
- (ii) Draw the scatter diagram for these values, labelling the axes clearly. Explain which of the following equations, where a and b are constants and $b > 0$, provides the most accurate model of the relationship between t and w .
- (A) $w = a + b \ln t$
- (B) $w = a + bt^2$
- (C) $w = a + \frac{b}{t^2}$
- [2]
- (iii) Using the model you chose in part (ii), write down the equation for the relationship between t and w , giving the numerical values of the coefficients. State the product moment correlation coefficient for this model and comment on its value. [3]
- (iv) Given that a metal plate being immersed in the strong acid solution for t hours has a weight loss of 2.4%, estimate the value of t . Give two reasons why this estimate is reliable. [3]
- (v) Given that 1 day = 24 hours, re-write your equation from part (iii) so that it can be used to estimate the percentage weight loss of metal plates when the length of time of immersing the metal plates in the strong acid solution is measured in days. [1]

- 11 (a)** Flash Electrics is a company which specializes in installing electricity meters in new houses in Central City. The time taken, T minutes, by its employees to install an electricity meter may be assumed to be normally distributed with mean μ and standard deviation σ .

(i) Given that $P(T < 40) = P(T > 50) = 0.36$, find the values of μ and σ . [3]

(ii) A random sample of 18 new houses in Central City with electricity meters installed by Flash Electrics is taken. Find the probability that at least 5 but fewer than 10 houses in this sample have an installation time of at most 50 minutes. [3]

- (b)** The electricity consumption, measured in kilowatt hour (kWh), of the households in Central City has a normal distribution with mean 520 and standard deviation 35.

(i) Find the probability that the electricity consumption of a randomly chosen household is more than 500 kWh. [1]

(ii) Two households in Central City are randomly chosen. Find the probability that both households each have electricity consumption of less than 500 kWh. [2]

(iii) The probability that the total electricity consumption of two randomly chosen households is less than 1000 kWh is denoted by p . Without calculating its value, explain why p will be greater than your answer to part **(ii)**. [1]

The electricity consumption of the households in Star City has a normal distribution with mean 475 kWh and standard deviation 25 kWh. It is known that electricity costs \$0.18 per kWh and \$0.15 per kWh in Central City and Star City respectively.

Let X represent the electricity bill of a randomly chosen household in Central City.

Let Y represent the electricity bill of a randomly chosen household in Star City.

(iv) Find $P(4Y - 3X > 7)$ and explain, in the context of this question, what your answer represents. [5]

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